SUBCONSCIOUS → EMOTION

Author: Ryan J. Park (Stark) · October 24, 2026

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SPINE

$$S(t) = \int |\varepsilon(\tau)| \cdot x(\tau) \cdot e^{-(t-\tau)/T} \, d\tau$$

$$e = h - h^*$$

$$m \cdot \ddot{e} + c \cdot \dot{e} + k \cdot e = F(t)$$

$$D = \|e\|$$

$$r = -\dot{D}$$

$$m = f(S)$$

WHAT CHANGED FROM VERSION 2

- Spring moved onto e (the displacement) instead of D (the distance). A distance can't go below zero, but a spring swings through zero. Now the weight swings through the set point and D stays positive.
- ε became |ε|. Being wrong high and being wrong low no longer cancel in the integral.
- Valence is now $r = -\dot{D}$, the continuous version, matching the rest of the spine. The step version $D(t) - D(t+1)$ is an approximation of it.
- "Energy = mass over time" is kept as a picture only. In physics, energy = mass × (distance/time)², so the equals sign doesn't hold.
- Timescale separation added. It's what makes $m = f(S)$ allowed inside a plain spring equation.
- Homeostatic reinforcement learning moved to EXISTED ALREADY.

NEW SYMBOLS

- decay curve → $e^{-(t-\tau)/T}$: each old entry counts less the longer ago it was
- half-life of the subconscious → T: small T = only recent exposure matters; big T = childhood still weighs
- damping → c: decay of the emotion in the moment (the swing dies out)

– displacement → e = h − h*: which way and how far I am from the set point

TWO DECAYS, TWO CLOCKS

– c: the emotion settles. Seconds to days. Already in the spring equation as c·ė.
– T: the memory fades. Months to a lifetime. In the integral as the e⋯ term.
– Timescale separation: T is so much slower than c that m is effectively constant during any single emotion. That's why a plain spring equation still works even though m changes over a lifetime.
– Since m = f(S), mass fades as S fades. Heaviness is not permanent; it drains at the rate the history fades. (Picture: "energy = mass over time." Picture only, not an equation.)

FULL DICTIONARY

– over time (piled up) → ∫ dt
– how fast it's changing now → d/dt
– how wrong I was → |ε| (size of prediction error, sign removed)
– exposure → x
– where the body is / wants to be → h / h*
– displacement → e = h − h*
– uncomfort → D = ‖e‖ (length of the string)
– heaviness → m (mass)
– pull back to normal → k (spring)
– settling → c (damping)
– the event → F(t)
– better/worse → r = −Ḋ (valence: the string getting shorter is positive)
– swing speed → ω = √(k/m) (heavy = slow, light = fast)
– how long the swing lasts → $e^{-ct/2m}$ (heavy = long)
– memory fade → T (half-life)
– two clocks far apart → timescale separation

PLAIN READ

1. Subconscious = exposure, weighted by how wrong I was, stacked over time, with old entries fading. No error, nothing written.
2. Uncomfort = the length of the string between where I am and where I want to be.
3. Emotion = a weight on a spring hit by an event. It can swing past the set point and back. Light weight: flinch, settles fast. Heavy weight: barely moves, swings

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slow and long.

4. Valence = whether the string is getting shorter or longer, not how long it is.
5. History sets the mass. Mass sets the swing. Both fade on the T clock.
6. Felt layer = turning r from a number into something felt. Unwritten.

OPEN

- f: how exactly S becomes m. Unwritten.
- Line 6: the operator that turns r into something felt. Unwritten by anyone (hard problem).

EXISTED ALREADY

Prediction error, integral, leaky integrator (the decay term), set point, mass-spring-damper, natural frequency, timescale separation, valence as drive reduction (Keramati & Gutkin 2014, homeostatic reinforcement learning: reward = drop in distance from set point).

MINE

The assembly: S → m → swing, felt layer on top, two decay clocks. (Not yet checked whether this exact chain has been published.)

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